Formal Fixed-Confidence Synthetic Comparison: VB-EGE versus Pareto-Specific BT-GLR Track-and-Stop

Synthetic experiment report

July 28, 2026

Abstract

We compare VB-EGE with a Pareto-specific heuristic extension of the Bradley–Terry generalized-likelihood-ratio Track-and-Stop method of Goldberger and Rudi. The experiment contains 36,400 algorithm runs, 18,200 per method, and exactly reuses the formal synthetic settings and repetition counts used for VB-EGE: fixed-confidence scaling, confidence-scaling quantiles, eight benchmark families, and Pareto-front-size ablation. Both methods identify the all-coordinate-strict Pareto set and are evaluated by mean stopping time, one standard error, stopping rate, and set error. The Pareto heuristic has smaller mean stopping time in 1 of 42 parameter cells; its geometric-mean stopping-time ratio to VB-EGE is 15.5. Observed set errors are 0 for VB-EGE and 0 for the Pareto heuristic. The latter has 19 runs that reach the predeclared 10^{18} cap without stopping, versus 0 for VB-EGE. It is an explicitly documented high-dimensional extension, not a transfer of the scalar top-k correctness or optimality theorem.

1 Protocol, target, and reported quantities

1.1 Observation model and Pareto convention

There are K arms and d objectives. Arm i has utility vector $\theta_i \in \mathbb{R}^d$. A query of unordered pair $\{i, j\}$ on objective r returns

$$Y \sim \mathrm{Bernoulli}(p_{ij,r}), \qquad p_{ij,r} = \sigma(\theta_{i,r} - \theta_{j,r}), \qquad \sigma(x) = \frac{1}{1+e^{-x}}.$$

We use the main algorithm's all-coordinate-strict convention:

$$j \succ i \quad \Longleftrightarrow \quad \theta_{j,r} > \theta_{i,r} \quad \text{for every } r \in [d].$$

The target is $\mathcal{P}(\theta) = \{i : \nexists j \neq i \text{ with } j \succ i\}$. This convention is used in instance generation, both stopping rules, and error evaluation. Equality is therefore not silently replaced by ordinary weak Pareto dominance.

1.2 Fixed-confidence endpoint and metrics

For target failure probability δ, a run stops at query count τ and returns $\widehat{\mathcal{P}}$. Its set error is

$$E = \mathbf{1}\{\widehat{\mathcal{P}} \neq \mathcal{P}(\theta)\}.$$

The primary ordinate in every result figure is the arithmetic mean $\bar{\tau} = n^{-1} \sum_u \tau_u$. The shaded ribbon is $\bar{\tau} \pm \mathrm{SE}(\bar{\tau})$, where

$$\mathrm{SE}(\bar{\tau}) = \frac{s_\tau}{\sqrt{n_{\mathrm{ind}}}}.$$

Here n_{ind} counts independent latent instances. When a benchmark has several observation replicates for one latent instance, we first average within instance and compute SE across instance means. Thus the ribbon does not mistreat repeated observations of one θ as independent instances. The benchmark bars use the same mean and SE definition; their darker overlays mark the interval $\bar{\tau} \pm \text{SE}$. If every independent unit in a cell stops at the same geometric phase, the empirical SE is exactly zero and the ribbon correctly collapses onto the mean line; no artificial visual width is added.

A run that does not stop before the predeclared maximum is recorded at $\tau_{\max} = 10^{18}$. Consequently, a cell containing capped runs reports the cap-coded mean

$$\bar{\tau}_{\text{cap}} = \frac{1}{n} \sum_{u} \min\{\tau_u, \tau_{\max}\}.$$

This is a lower bound on the uncapped mean stopping time, not an estimate that pretends the run stopped at the cap. Its SE only describes dispersion of these cap-coded observations. The stopping-rate table and annotations on the benchmark figure identify every affected cell.

1.3 Formal experiment bank

Table 1: Formal protocol counts. Counts are per algorithm.

Section	Cells	Repetitions per cell	Runs
Fixed-confidence scaling	12	500	6,000
Confidence-scaling quantiles	14	500 Sym.; 300 Arena	5,800
Fixed-confidence benchmark suite	8	500	4,000
Pareto-size ablation	8	300	2,400
Total	42	–	18,200

1.4 Instance families

The Symmetric generator places one arm at the zero vector and every other arm at $-\Delta\mathbf{1}$, then randomly permutes arm identities; hence the true Pareto size is one. Arena instances draw s mutually non-dominated anchors from a rescaled Dirichlet trade-off surface and generate every remaining arm by subtracting an independent coordinate-wise margin from one anchor. Convex-2D uses 15 evenly spaced points on a decreasing line segment as its frontier; Convex-3D uses 15 Dirichlet trade-off anchors. Witness instances give each frontier arm four generated arms for which it is the unique frontier dominator. Two-group-10 draws 48 "low" arms independently from $[0.20, 0.45]^{10}$ and 16 "high" arms from $[0.55, 0.75]^{10}$; its true frontier size is computed from the draw rather than imposed.

Table 2: Exact benchmark-suite generator settings. All use $\delta = 0.05$ and 500 algorithm runs.

Setting	(K, d, s) or group sizes	Generator	Additional parameters
Convex-2D	$(60, 2, 15)$	convex frontier	margins $[0.03, 0.18]$
Convex-3D	$(60, 3, 15)$	Dirichlet frontier	$\alpha = 1$, margins $[0.03, 0.18]$
Arena-4 small	$(32, 4, 8)$	trade-off arena	$\alpha = 0.7$, margins $[0.08, 0.25]$
Arena-4 medium	$(64, 4, 12)$	trade-off arena	$\alpha = 0.7$, margins $[0.08, 0.25]$
Arena-10 medium	$(64, 10, 16)$	trade-off arena	$\alpha = 0.7$, margins $[0.06, 0.20]$
Witness-4	$(40, 4, 8)$	unique witness	$q/p = 4$, margins $[0.04, 0.16]$
Witness-10	$(80, 10, 16)$	unique witness	$q/p = 4$, margins $[0.035, 0.14]$
Two-group-10	$(48 + 16, 10, \text{drawn})$	high/low groups	intervals stated above

Convex-3D and Arena-10 medium use banks of 50 independently generated latent instances with 10 observation replicates per instance. Every other benchmark uses 500 independently generated latent instances. This distinction is why their reported SEs are clustered by latent instance.

All paired rows share the same latent instance θ but use independent, deterministic observation seeds. There is no result-dependent filtering and no per-setting retuning. Fixed-confidence scaling and all benchmark/ablation cells use $\delta = 0.05$. Confidence scaling varies δ while reusing the same latent instance within a replication.

2 Algorithms

2.1 VB-EGE

VB-EGE samples a focal arm i and objective r against a uniformly random opponent. The resulting coordinate-wise Vector-Borda target is

$$b_{i,r} = \frac{1}{K-1}\sum_{j\neq i}\sigma(\theta_{i,r} - \theta_{j,r}).$$

At phase m, it uses $\varepsilon_m = 2^{-m}$,

$$L_m = \log\left(\frac{c_{\log} K d m^2}{\delta}\right), \qquad n_m = \left\lceil \frac{c_s L_m}{\varepsilon_m^2} \right\rceil,$$

and compares empirical accept/reject gaps with $c_\theta\sqrt{L_m/(2n_m)}$. The practical constants $(c_s, c_\theta, c_{\log}) = (2, 4, 4)$ are fixed in every cell. These practical constants differ from conservative proof constants but were selected before the experiment, checked by the separate constant-sensitivity study, and then held fixed throughout all sections.

2.2 Scalar BT-GLR Track-and-Stop source

Goldberger and Rudi [1] study scalar top-k identification. Their algorithm fits a constrained Bradley–Terry MLE, profiles the likelihood over alternatives that reverse a top-k boundary, tracks a max–min information allocation, forces exploration, and stops when a GLR statistic exceeds a time-uniform mixture threshold. The theorem is for the scalar top-k alternative geometry. A Pareto frontier is not characterized by one inside/outside boundary pair, so that theorem does not directly specify a d-objective algorithm.

2.3 Pareto-specific extension used here

Coordinate-wise BT MLE. For each objective r, all sampled opponent identities are retained and an independent centered, box-constrained BT MLE is fitted:

$$\widehat{\theta}_r \in \arg\max_{\vartheta} \ell_{r,t}(\vartheta) \quad \text{s.t.} \quad \mathbf{1}^\top\vartheta = 0, \qquad \|\vartheta\|_\infty \le B, \qquad B = 2.$$

This is the same pair-objective transcript used by the Track-and-Stop sampler; it is not a Borda reduction. The box contains every centered formal instance and prevents complete-separation non-existence.

Frontier-changing alternatives. Let $\widehat{P} = \mathcal{P}(\widehat{\theta})$. In the closure of the strict alternative set, removing $p \in \widehat{P}$ can be witnessed by another current frontier arm p' satisfying

$$\lambda_{p',r} \ge \lambda_{p,r} \quad \forall r.$$

Adding $q \notin \widehat{P}$ requires breaking every current frontier arm as a possible dominator:

$$\forall p \in \widehat{P}, \quad \exists r \in [d] \text{ such that } \lambda_{q,r} \geq \lambda_{p,r}.$$

Equivalently, an add branch is a union over $d^{|\widehat{P}|}$ witness assignments $a : \widehat{P} \to [d]$. This remove/add decomposition follows the Pareto-front identification geometry used by Crépon, Garivier, and Koolen [2].

Exact small path and formal large-front path. The implementation contains an exact certificate-enumeration path for $K \leq 8$ when all add assignments fit under the enumeration cap. It profiles the original BT likelihood under each grouped set of order constraints and uses numerical primal–dual bounds. That path is covered by unit and smoke tests, but none of the present formal cells uses it: all have $K \geq 16$.

For formal-size fronts, we first compute local-Fisher order costs

$$q_{ab,r} = \frac{[\widehat{\theta}_{b,r} - \widehat{\theta}_{a,r}]_+^2}{2(e_a - e_b)^\top H_r^+(e_a - e_b)},$$

where H_r is the observed BT Fisher Laplacian. Every remove branch receives screening score $\sum_r q_{p'p,r}$. Each add arm receives

$$s_q = \max_{p \in \widehat{P}} \min_{r \in [d]} q_{qp,r}.$$

The globally smallest screening branch is selected. For an add arm, a feasible joint witness assignment is built from the coordinate-wise cheapest assignments, all same-coordinate assignments, and retained assignments from earlier phases. The original constrained BT likelihood is then optimized for that one joint certificate. Denote its numerical profile lower value by G_t^{cert} and the global screening value by S_t. The implemented stopping statistic is

$$Z_t = \min\{S_t, G_t^{\text{cert}}\}.$$

This is substantially closer to a Pareto profile likelihood than the earlier single-pair local-quadratic prototype: the active branch is a jointly feasible frontier-changing certificate and is evaluated with the original BT likelihood. However, S_t is a scalable screening heuristic rather than a proved lower bound on the global Pareto profile. Therefore the formal large-front statistic is explicitly marked *not certified*.

Allocation and threshold. The fitted MLE and active feasible alternative define one-cell information

$$I_{ij,r} = \mathrm{kl}\Big(\sigma(\widehat{\theta}_{i,r} - \widehat{\theta}_{j,r}), \sigma(\widetilde{\theta}_{i,r} - \widetilde{\theta}_{j,r})\Big).$$

Exponentiated mirror updates track high-information cells; C-tracking converts the target proportions into integer queries. Uniform mass $\rho_t = t^{-1/3}$ is mixed over all $dK(K-1)/2$ pair-objective cells, and geometric checks use target size $\lceil 1.8t \rceil$. The stopping threshold is

$$\beta_t(\delta) = \log\frac{1}{\delta} + \frac{\lambda}{2}\sum_r \|\widehat{\theta}_r\|_2^2 + \frac{1}{2}\sum_r \log\det\Big(I + \frac{\sigma^2}{\lambda}L_{r,t}\Big),$$

with $(\lambda, \sigma^2) = (0.1, 0.25)$ and count Laplacian $L_{r,t}$. The run stops when all MLE/profile fits converge and $Z_t \geq \beta_t(\delta)$.

Status of the method. This report tests a fixed-confidence *heuristic extension*: it uses δ in a time-uniform-threshold-shaped stopping rule and empirically audits errors, but it does not claim that the scalar ICML theorem proves δ-correctness or asymptotic optimality after the Pareto extension. That limitation is methodological, not hidden in a footnote.

Table 3: Pareto BT-GLR implementation constants, fixed before the formal run.

Parameter	Value	Parameter	Value
Box bound B	2	Burn-in per pair-objective cell	2
Mixture λ	0.1	Sub-Gaussian proxy σ^2	0.25
Batch growth	1.8	Mirror step	0.8
Forced-exploration exponent	1/3	Maximum queries	10^{18}
Exact-profile arm cutoff	8	Screened certificate pool	8
MLE tolerance	10^{-7}	Maximum MLE iterations	1,000

3 Fixed-confidence scaling

This section fixes $\delta = 0.05$ and varies one structural parameter at a time: $K \in \{16, 32, 64, 128\}$ at $(d, \Delta) = (4, 1)$; $d \in \{2, 4, 10\}$ at $(K, \Delta) = (64, 1)$; and $\Delta \in \{0.5, 0.75, 1, 1.25, 1.5\}$ at $(K, d) = (64, 10)$. The horizontal axes are respectively arm count, objective count, and latent separation; the vertical axis is mean stopping time on a logarithmic scale.

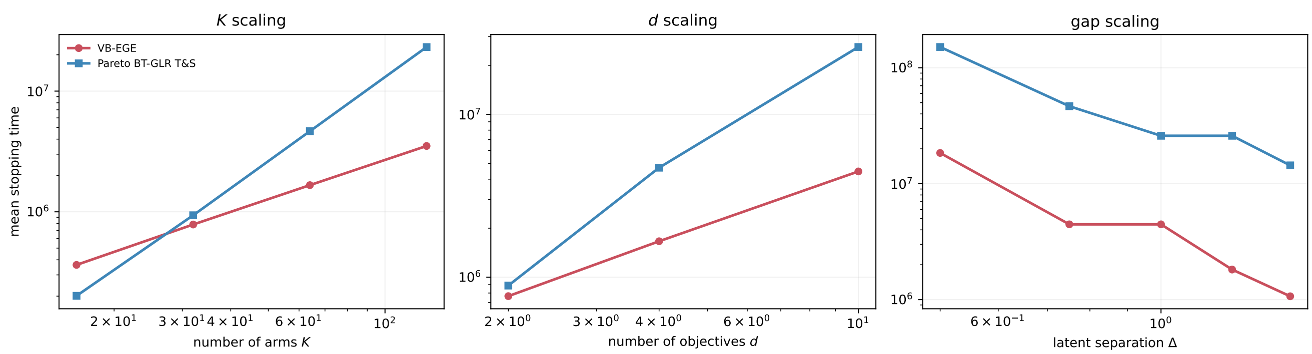

Figure 1: Fixed-confidence scaling. Lines show mean stopping time and shaded ribbons show mean $\pm$ one standard error. Lower is better.

Table 4: Fixed-confidence scaling at $\delta = 0.05$. Means and standard errors are computed from independent latent-instance units.

Cell	n	VB-EGE mean $\pm$ SE	Pareto mean $\pm$ SE	τ_P/τ_V	Errors V;P
$K = 16$	500	$3.63\times10^5 \pm 6.78\times10^2$	$2.01\times10^5 \pm 1.74\times10^3$	0.554	0/500;0/500
$K = 32$	500	$7.85\times10^5 \pm 4.11\times10^2$	$9.36\times10^5 \pm 1.20\times10^4$	1.19	0/500;0/500
$K = 64$	500	$1.66\times10^6 \pm 4.57\times10^2$	$4.65\times10^6 \pm 5.68\times10^4$	2.8	0/500;0/500
$K = 128$	500	$3.51\times10^6 \pm 3.01\times10^2$	$2.32\times10^7 \pm 2.92\times10^4$	6.6	0/500;0/500
$d = 2$	500	$7.65\times10^5 \pm 1.63\times10^3$	$8.89\times10^5 \pm 0$	1.16	0/500;0/500
$d = 4$	500	$1.66\times10^6 \pm 4.05\times10^2$	$4.70\times10^6 \pm 5.64\times10^4$	2.83	0/500;0/500
$d = 10$	500	$4.46\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	5.81	0/500;0/500
$\Delta = 0.5$	500	$1.84\times10^7 \pm 2.78\times10^3$	$1.51\times10^8 \pm 0$	8.2	0/500;0/500
$\Delta = 0.75$	500	$4.46\times10^6 \pm 0$	$4.66\times10^7 \pm 0$	10.5	0/500;0/500
$\Delta = 1$	500	$4.46\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	5.81	0/500;0/500
$\Delta = 1.25$	500	$1.82\times10^6 \pm 2.04\times10^4$	$2.59\times10^7 \pm 0$	14.3	0/500;0/500
$\Delta = 1.5$	500	$1.07\times10^6 \pm 2.12\times10^2$	$1.44\times10^7 \pm 0$	13.5	0/500;0/500

Result. The Pareto BT-GLR heuristic has the smaller mean stopping time in 1 of 12 cells. Its Pareto/VB-EGE mean ratio ranges from 0.554 ($K = 16$) to 14.3 ($\Delta = 1.25$), with geometric mean 4.1 across the section. This is a finite-protocol comparison; it is not an optimality claim.

4 Confidence-scaling quantiles

The horizontal coordinate is $\log(1/\delta)$ and the ordinate is mean stopping time. Symmetric $K = 64, d = 10$ uses $\delta \in \{0.2, 0.1, 0.05, 0.02, 0.01, 0.005, 0.002, 0.001\}$ with 500 paired replications. Arena-10 medium uses the first six values through 0.005 with 300 paired replications. Reusing each latent θ across confidence levels isolates the effect of the requested confidence from instance variation.

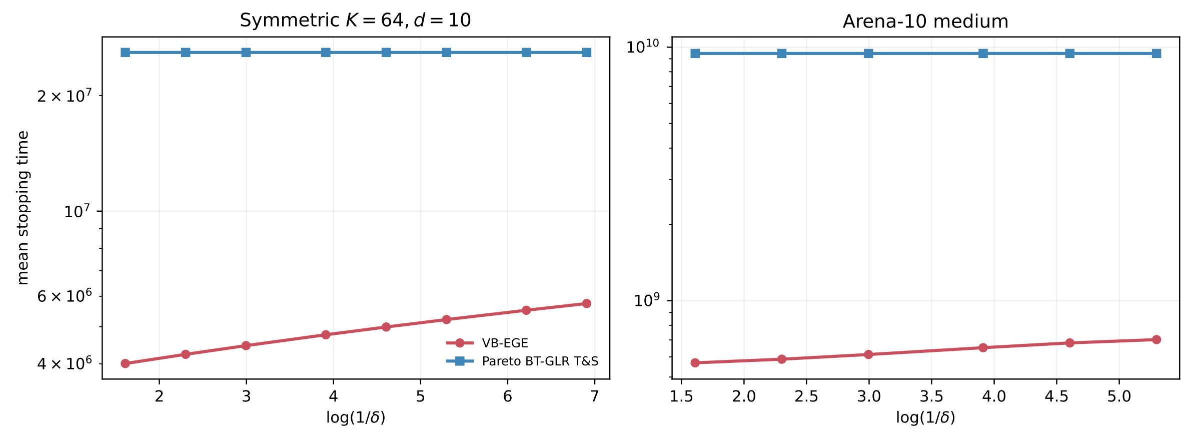

Figure 2: Confidence scaling. Lines are mean stopping time and ribbons are one standard error. The abscissa is $\log(1/\delta)$, not δ.

Table 5: Confidence-scaling quantiles. The same latent replication is reused across δ values; each displayed SE is computed within its fixed-δ cell.

Cell	n	VB-EGE mean ± SE	Pareto mean ± SE	τ_P/τ_V	Errors V;P
Sym., $\delta = 0.2$	500	$4.01\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	6.47	0/500;0/500
Sym., $\delta = 0.1$	500	$4.23\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	6.12	0/500;0/500
Sym., $\delta = 0.05$	500	$4.46\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	5.81	0/500;0/500
Sym., $\delta = 0.02$	500	$4.76\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	5.44	0/500;0/500
Sym., $\delta = 0.01$	500	$4.99\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	5.19	0/500;0/500
Sym., $\delta = 0.005$	500	$5.22\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	4.97	0/500;0/500
Sym., $\delta = 0.002$	500	$5.52\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	4.7	0/500;0/500
Sym., $\delta = 0.001$	500	$5.74\times10^6 \pm 0$	$2.59\times10^7 \pm 0$	4.51	0/500;0/500
Arena-10, $\delta = 0.2$	300	$5.69\times10^8 \pm 3.51\times10^6$	$9.44\times10^9 \pm 8.79\times10^7$	16.6	0/300;0/300
Arena-10, $\delta = 0.1$	300	$5.88\times10^8 \pm 3.51\times10^6$	$9.44\times10^9 \pm 8.79\times10^7$	16.1	0/300;0/300
Arena-10, $\delta = 0.05$	300	$6.13\times10^8 \pm 3.64\times10^6$	$9.44\times10^9 \pm 8.79\times10^7$	15.4	0/300;0/300
Arena-10, $\delta = 0.02$	300	$6.53\times10^8 \pm 4.19\times10^6$	$9.44\times10^9 \pm 8.79\times10^7$	14.5	0/300;0/300
Arena-10, $\delta = 0.01$	300	$6.82\times10^8 \pm 4.28\times10^6$	$9.44\times10^9 \pm 8.79\times10^7$	13.9	0/300;0/300
Arena-10, $\delta = 0.005$	300	$7.02\times10^8 \pm 4.39\times10^6$	$9.44\times10^9 \pm 8.79\times10^7$	13.4	0/300;0/300

Table 6: Descriptive fits $\bar{\tau} = a + b\log(1/\delta)$. The slope is not a claimed asymptotic constant for the Pareto heuristic. R^2 is undefined, shown as –, when the response is numerically constant.

Setting	Method	Slope b	R^2
Symmetric	VB-EGE	3.28×10^5	1.0000
Symmetric	Pareto BT-GLR T&S	0	–
Arena-10	VB-EGE	3.77×10^7	0.9959
Arena-10	Pareto BT-GLR T&S	0	–

Result. The Pareto BT-GLR heuristic has the smaller mean stopping time in 0 of 14 cells. Its Pareto/VB-EGE mean ratio ranges from 4.51 (Sym., $\delta = 0.001$) to 16.6 (Arena-10, $\delta = 0.2$), with geometric mean 8.32 across the section. This is a finite-protocol comparison; it is not an optimality claim.

5 Fixed-confidence benchmark suite

The eight settings probe distinct frontier geometries rather than one smooth scaling path: Convex-2D, Convex-3D, Arena-4 small, Arena-4 medium, Arena-10 medium, Witness-4, Witness-10, and Two-group-10. The vertical axis is mean stopping time at $\delta = 0.05$ on a logarithmic scale. The darker interval inside each pastel bar is mean ± one SE.

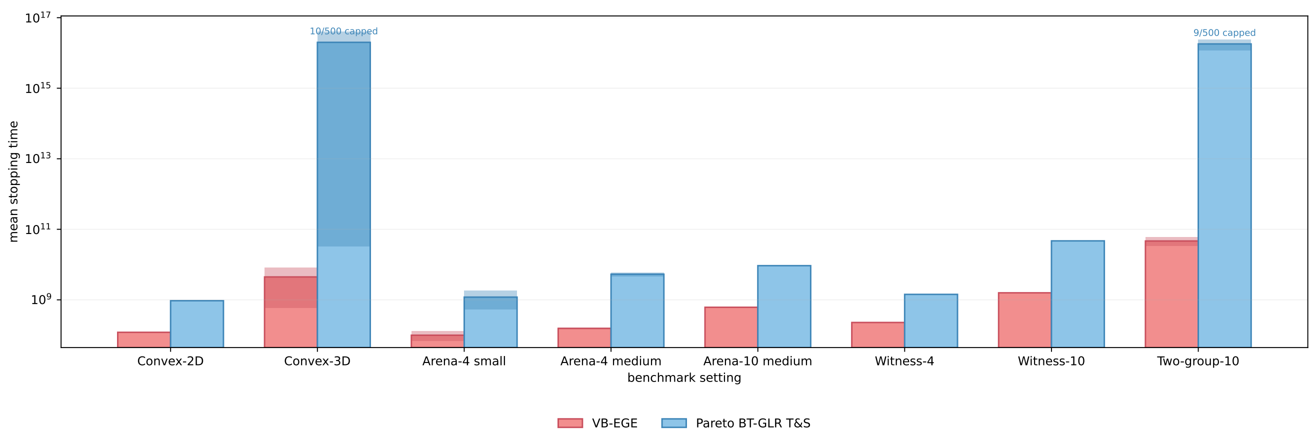

Figure 3: All eight fixed-confidence benchmarks in one panel. Lower bars are better; darker overlays show mean ± one standard error. Text above a bar reports runs that reached the 10^{18} cap without stopping. Means in those cells are cap-coded lower bounds on uncapped mean stopping time.

Table 7: Eight fixed-confidence benchmark settings. For instance banks with multiple observation replicates, SE is clustered by latent instance.

Cell	n	VB-EGE mean ± SE	Pareto mean ± SE	τ_P/τ_V	Errors V;P
Convex-2D	500	$1.20\times10^8 \pm 9.06\times10^5$	$9.44\times10^8 \pm 9.47\times10^6$	7.87	0/500;0/500
Convex-3D	500	$4.46\times10^9 \pm 3.86\times10^9$	$2.00\times10^{16} \pm 2.00\times10^{16}$	4.49e+06	0/500;0/490
Arena-4 small	500	$9.93\times10^7 \pm 3.11\times10^7$	$1.19\times10^9 \pm 6.53\times10^8$	12	0/500;0/500
Arena-4 medium	500	$1.55\times10^8 \pm 5.90\times10^6$	$5.33\times10^9 \pm 6.92\times10^8$	34.5	0/500;0/500
Arena-10 medium	500	$6.15\times10^8 \pm 5.87\times10^6$	$9.31\times10^9 \pm 5.92\times10^7$	15.1	0/500;0/500
Witness-4	500	$2.28\times10^8 \pm 7.80\times10^5$	$1.43\times10^9 \pm 2.20\times10^6$	6.27	0/500;0/500
Witness-10	500	$1.58\times10^9 \pm 2.20\times10^6$	$4.70\times10^{10} \pm 0$	29.7	0/500;0/500
Two-group-10	500	$4.66\times10^{10} \pm 1.32\times10^{10}$	$1.80\times10^{16} \pm 5.95\times10^{15}$	3.87e+05	0/500;0/491

Result. The Pareto BT-GLR heuristic has the smaller mean stopping time in 0 of 8 cells. Its Pareto/VB-EGE mean ratio ranges from 6.27 (Witness-4) to 4.49e+06 (Convex-3D), with geometric mean 251 across the section. This is a finite-protocol comparison; it is not an optimality claim.

Pareto BT-GLR reaches the cap in 10 of 500 Convex-3D runs and 9 of 500 Two-group-10 runs. All other benchmark runs stop. The corresponding displayed Pareto/VB-EGE ratios are therefore conservative lower bounds: removing the cap cannot turn either cell into a Pareto BT-GLR win.

6 Pareto-size ablation

This section fixes $K = 64$ and varies the true frontier size $|P| \in \{4, 8, 16, 32\}$ for $d = 4$ and $d = 10$. The abscissa is the generator's verified true Pareto-set size, and the ordinate is mean stopping time at $\delta = 0.05$. This experiment directly tests the combinatorial pressure in the add-frontier alternative.

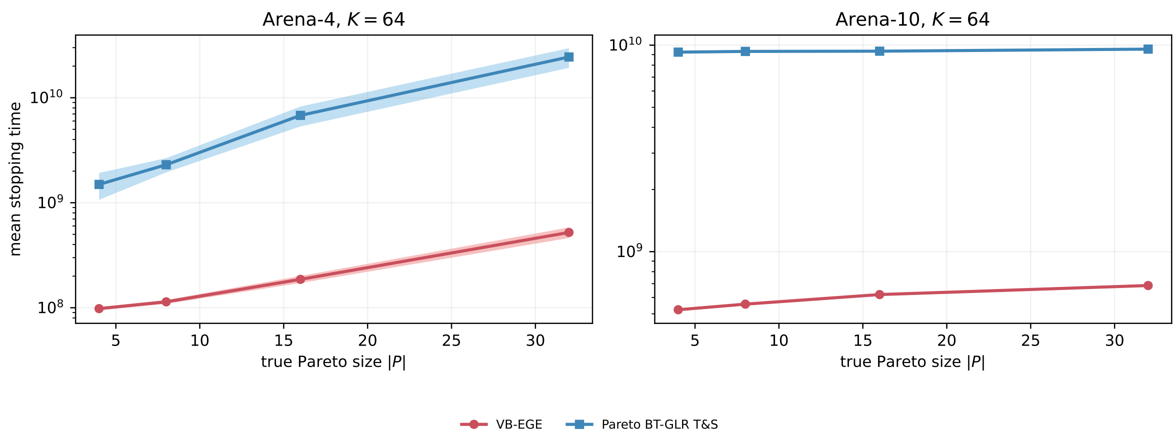

Figure 4: Pareto-size ablation. Lines show mean stopping time and ribbons show mean ± one standard error.

Table 8: Pareto-front-size ablation at $K = 64$. Means and one standard error use latent instances as independent units.

Cell	n	VB-EGE mean ± SE	Pareto mean ± SE	τ_P/τ_V	Errors V;P
$d = 4,\ \lvert P\rvert = 4$	300	$9.81\times10^7 \pm 3.40\times10^6$	$1.49\times10^9 \pm 4.28\times10^8$	15.2	0/300;0/300
$d = 4,\ \lvert P\rvert = 8$	300	$1.14\times10^8 \pm 4.85\times10^6$	$2.30\times10^9 \pm 3.57\times10^8$	20.2	0/300;0/300
$d = 4,\ \lvert P\rvert = 16$	300	$1.86\times10^8 \pm 1.45\times10^7$	$6.79\times10^9 \pm 1.46\times10^9$	36.4	0/300;0/300
$d = 4,\ \lvert P\rvert = 32$	300	$5.21\times10^8 \pm 5.98\times10^7$	$2.44\times10^{10} \pm 5.21\times10^9$	46.9	0/300;0/300
$d = 10,\ \lvert P\rvert = 4$	300	$5.23\times10^8 \pm 2.90\times10^6$	$9.25\times10^9 \pm 0$	17.7	0/300;0/300
$d = 10,\ \lvert P\rvert = 8$	300	$5.57\times10^8 \pm 3.14\times10^6$	$9.33\times10^9 \pm 4.26\times10^7$	16.7	0/300;0/300
$d = 10,\ \lvert P\rvert = 16$	300	$6.20\times10^8 \pm 3.82\times10^6$	$9.35\times10^9 \pm 4.91\times10^7$	15.1	0/300;0/300
$d = 10,\ \lvert P\rvert = 32$	300	$6.85\times10^8 \pm 6.43\times10^6$	$9.57\times10^9 \pm 1.03\times10^8$	14	0/300;0/300

Result. The Pareto BT-GLR heuristic has the smaller mean stopping time in 0 of 8 cells. Its Pareto/VB-EGE mean ratio ranges from 14 ($d = 10,\ |P| = 32$) to 46.9 ($d = 4,\ |P| = 32$), with geometric mean 20.6 across the section. This is a finite-protocol comparison; it is not an optimality claim.

7 Reliability and implementation diagnostics

Table 9: Stopping and empirical-error audit. Wilson upper is the two-sided 95% Wilson upper endpoint for the conditional error rate among stopped runs.

Section	Method	n	Stop rate	Errors	Wilson upper	MLE conv.
Scaling	VB-EGE	6000	1.0000	0/6000	0.0006	1.0000
Scaling	Pareto BT-GLR	6000	1.0000	0/6000	0.0006	1.0000
Confidence	VB-EGE	5800	1.0000	0/5800	0.0007	1.0000
Confidence	Pareto BT-GLR	5800	1.0000	0/5800	0.0007	1.0000
Benchmarks	VB-EGE	4000	1.0000	0/4000	0.0010	1.0000
Benchmarks	Pareto BT-GLR	4000	0.9952	0/3981	0.0010	1.0000
Pareto size	VB-EGE	2400	1.0000	0/2400	0.0016	1.0000
Pareto size	Pareto BT-GLR	2400	1.0000	0/2400	0.0016	1.0000

Across cells, the minimum stopping rates are 1.0000 for VB-EGE and 0.9800 for Pareto BT-GLR. The minimum Pareto BT-GLR MLE convergence rate is 1.0000. A zero observed error count is computed only among stopped runs and is not a proof of zero risk; Table 9 reports Wilson upper endpoints to make the finite-repetition resolution explicit. In particular, the 19 capped Pareto BT-GLR benchmark runs have no returned set and are excluded from the conditional set-error denominator, while remaining visible through the stop rate. At the individual-cell level, zero failures in 500 repetitions has a two-sided 95% Wilson upper endpoint of approximately 0.0076, and zero in 300 has endpoint approximately 0.0126. Consequently these repetitions can reveal gross miscalibration but cannot empirically certify a target such as $\delta = 0.001$.

All formal Pareto BT-GLR rows use the large-front profile mode recorded as `screened-joint-certificate`. The fraction marked as theorem-level certified statistics is 0.0000, as expected from the declared heuristic status. The implementation nevertheless checks numerical convergence, uses a joint feasible certificate for the active branch, records the original-BT profile estimate and bounds, and records the threshold at stopping.

8 Conclusions

The experiment supports a comparison between two fully specified fixed-confidence stopping procedures under the same formal synthetic bank. It does not support relabeling the Pareto extension as the original scalar ICML algorithm. VB-EGE operates in Vector-Borda certificate space; Pareto BT-GLR fits the full coordinate-wise BT model and adaptively targets a frontier-changing certificate. Their stopping-time difference therefore combines estimator dimension, alternative geometry, sampling allocation, and stopping calibration.

Numerically, Pareto BT-GLR wins 1 of 42 aggregate-mean cells and has a cross-cell geometric mean ratio of 15.5 relative to VB-EGE. The section-level figures and tables identify where that advantage or disadvantage occurs; the empirical error audit states the resolution supported by the actual repetition counts.

9 Reproducibility

```
python -u -m icml_pareto_track_stop.run_formal \
```

```
  --jobs 16 --prepare-jobs 1 --checkpoint-every 25 --resume
python -m icml_pareto_track_stop.summarize_formal
python -m icml_pareto_track_stop.report.build_certificate_v1_formal_report
```

The run-level CSV, resumable JSONL checkpoint, aggregate summaries, paired summaries, diagnostics, figures, TeX source, and PDF all reside under `icml_pareto_track_stop/`. Generated data and build artifacts are kept out of version control; only the canonical report PDF is published.

References

[1] M. Goldberger and N. Rudi. Optimal Top-k Identification from Pairwise Comparisons. *Proceedings of the 43rd International Conference on Machine Learning*, PMLR 306, 2026. https://arxiv.org/abs/2607.08979.

[2] B. Crépon, A. Garivier, and W. M. Koolen. Sequential Learning of the Pareto Front for Multi-objective Bandits. *Proceedings of AISTATS*, PMLR 238, 2024. https://proceedings.mlr.press/v238/crepon24a.html.